Vision-language-action policies versus world models for robot learning

What each family predicts, how the prediction reaches the actuator, and how far the evidence actually goes

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1. Short answer for a group entering the field

Both families are trained on robot trajectories with supervised or self-supervised objectives; the difference that matters is *what the target is* and *when the target is consumed*. A vision-language-action (VLA) policy predicts actions: the observation and an instruction go in, a chunk of low-level commands comes out, and the chunk is executed immediately. A world model predicts future observations (pixels, latents, or a reward), and that prediction is used in one of three ways: inside the control loop, as the thing being optimised; offline, to manufacture extra training data for a conventional policy; or inside a learned environment, where a policy is trained by reinforcement on imagined rollouts.

If the practical question is "which gives me a working manipulator soonest", the published record currently favours the first family: VLA-style policies are the systems that have been exercised on real hardware at several institutions, with blind or randomised evaluation protocols. World models have the more interesting scientific question and, so far, the narrower evidence: most claims of large gains are single-site, and several are measured in simulation or in games. This report tries to make that asymmetry visible rather than to declare a winner.

Two cautions frame everything below. First, a predicted video that looks plausible is not a control result: pixel fidelity and closed-loop task success are different quantities, and one does not imply the other. Second, the numbers in the cited papers are produced by different data, embodiments, task sets and evaluation protocols, and are reported here as the authors' own claims under their own conditions, not as points on a common scale.

2. Scope, dates and evidence base

This is a small, opportunistically assembled corpus, not a systematic review. Sixteen unique papers are cited. Their centre of mass is 2024–2025: OpenVLA, pi-0, DINO-WM [Kim et al., 2024, Black et al., 2024, Zhou et al., 2024] and the LIBERO benchmark [Liu et al., 2023] form the 2023–2024 layer; SmolVLA, GR00T N1, Gemini Robotics 1.5, WorldVLA, V-JEPA 2, DreamGen, Cosmos, Dreamer 4, RoboArena and the large behaviour-model study [Shukor et al., 2025, NVIDIA, 2025c, Gemini Robotics Team, Google DeepMind, 2025, Cen et al., 2025, Meta FAIR, 2025, NVIDIA, 2025b,a, Hafner et al., 2025, RoboArena, 2025, LBM blind-trial study, 2025] supply the 2025 layer.

The 2026 layer is thin and must be labelled as such. Exactly two 2026 items are discussed: World-Gymnast, which we read as a complete text, and PlayWorld, for which a full-text HTML acquisition is retained in our evidence bundle but from which this report used only abstract-level statements, so it is described strictly at that level [World-Gymnast, 2026, PlayWorld, 2026]. A search cutoff of 19 October 2026 therefore does *not* mean complete coverage of 2026; it means that the 2026 slice of this corpus is two papers. Whenever this report says that something was "not located", the claim is about these sixteen papers only. It is not a statement that no such result exists in the literature.

What was actually done, and what was not: sources were acquired as HTML or PDF full text where available, and claims about them are checked against those texts rather than against abstracts. Nothing was re-implemented, no benchmark was re-run, and no new measurement was made. All numbers below are the papers' own, quoted with their conditions. Coverage of individual papers is uneven and is stated where it matters: for a few items

only metadata or abstract-level text was used, and at least one of those (PlayWorld) was in fact acquired as a substantial full-text HTML bundle that this report did not read for its body.

3. The two axes that actually separate the families

Discussions of "VLA versus world model" usually blur two independent choices. The first is what the network is trained to output: actions, future observations, or both. The second is where that output is consumed: in the control loop, as training data, or as a training environment. Table 1 arranges the families along those two axes. Figure 1 draws the same information as data flow.

A third axis, cost, follows from the second. Direct action policies pay once per executed chunk: a single forward pass, or a short fixed number of denoising steps for a flow-matching head, followed by open-loop execution for a few steps. Planning with a world model pays per candidate sequence, because the model must be rolled forward for each sample and usually for several refinement iterations. This is the dominant practical difference between the families at deployment time, and it is independent of how good either model is.

Family	Trained to predict	How prediction is used	Supervision that carries the load	Typical inference cost
A. Direct action policy (VLA)	action chunk	executed as the controller	action labels from teleoperation or mixed robot data	one pass per chunk, or a few denoising steps for a flow head
B. Action model with auxiliary future	actions **and** future frames	actions executed; frames are an auxiliary target	actions plus video, jointly	one pass per chunk for autoregressive heads
C. World model as planner	future latent or frames under candidate actions	optimised online (sampling, gradient or RL) then executed	action-conditioned video, sometimes action-free	many rollouts per control step
D. World model as data or trainer	future frames, or reward and value inside imagination	offline: pseudo-labels, data augmentation, or RL inside the model	video plus inverse-dynamics or reward labels	zero at deployment for D1; amortised for D2

Table 1. Four families on the two axes that matter. The rows describe the dominant pattern of each family, not every implementation: within family A, for instance, pi-0 uses an iterative flow-matching head rather than a single regression pass, and re-planning frequency varies by system [Black et al., 2024].

4. Family A: policies that predict actions and nothing else

OpenVLA is the cleanest reference point: a 7B-parameter vision-language model fine-tuned to emit discretised action tokens, trained on roughly 970k robot episodes drawn from the Open X-Embodiment mixture, and evaluated on a Franka arm across a set of language-conditioned manipulation tasks [Kim et al., 2024]. pi-0 replaces the discrete decoder with a flow-matching action expert attached to a pretrained VLM and reports training on the order of 10,000 hours of robot data spanning seven robot configurations and 68 tasks, with control at up to 50 Hz [Black et al., 2024]. SmolVLA pushes the other way: a 450M-parameter policy trained on 481 community datasets (about 22.9k episodes and 10.6M frames), designed to train on a single GPU and to run asynchronous inference on consumer hardware [Shukor et al., 2025]. GR00T N1 is a dual-system design in which a vision-language component supplies the slower reasoning and a diffusion-transformer component emits the action chunk, trained on a data pyramid that mixes web video, synthetic and real robot data, with a humanoid target [NVIDIA, 2025c]. Gemini Robotics 1.5 adds an explicit "think before acting" stage and reports transferring motion between embodiments [Gemini Robotics Team, Google DeepMind, 2025].

The common property is that no future observation is ever a training target. Every gradient comes from an action or from the language grounding that supports it. This is why the family is cheap to deploy and why it inherits the

weaknesses of behaviour cloning: distribution shift at test time is unmodelled, and a chunk executed open-loop cannot notice that the world has diverged from the states in the training set.

The evidence for this family is the strongest in the corpus, but it is uneven in a specific way. Single-site real robot results are routine. Multi-institution evaluation is rare, and the one systematic instance here is RoboArena, which distributes pairwise comparisons of generalist policies over a shared robot pool and reports that rankings stabilise after roughly a hundred pairwise comparisons [RoboArena, 2025]. The most informative single study is the large behaviour-model examination, which ran about 1,800 blind rollouts across many tasks on a bimanual platform after training on roughly 1,700 hours of data, and reports data-efficiency advantages for large behaviour models over smaller baselines [LBM blind-trial study, 2025]. Those are the results that make the family credible; they are also, deliberately, results about *one* platform family and one evaluation protocol.

5. Family B: action models with auxiliary future prediction

WorldVLA sits between the families: a single autoregressive transformer over interleaved action and image tokens, trained so that the same model both predicts the next action and predicts the next frame [Cen et al., 2025]. The authors report an improvement of about 4% on LIBERO task suites over comparable action-only baselines, together with a reported 10% improvement in video-generation FVD; in their analysis the auxiliary image objective acts as a regulariser that stabilises the action objective. Crucially, only the action tokens reach the robot. The predicted frames are a co-training signal, not a plan.

This family is worth understanding because it isolates the value of prediction with everything else held fixed: the architecture, data and evaluation stay close to family A, and only the auxiliary target changes. The honest reading of the reported gains is that they are modest and measured largely in simulation, and that the benefit is described by the authors themselves as a training-time effect rather than a planning capability. A reader who wants to test "does predicting the future help control" cheaply will find this the most tractable experiment in the whole area, because the baseline is the same model minus one loss term.

6. Family C: world models that plan

Here the prediction is the controller. DINO-WM builds its world model on frozen DINOv2 features rather than on pixels, trains it on offline play data, and plans by rolling the latent model forward under sampled action sequences and scoring them against a goal image [Zhou et al., 2024]. The paper reports zero-shot planning across six environments, including manipulation and navigation, and argues that pre-trained visual features make the dynamics model learnable from modest data. V-JEPA 2-AC takes the same latent-prediction philosophy at larger scale: a video-pretrained encoder, a small action-conditioned predictor trained on under 62 hours of unlabelled DROID interaction video, and goal-conditioned planning by cross-entropy-method search in latent space, with real-robot pick-and-place demonstrations on two robot platforms in two laboratories [Meta FAIR, 2025].

Two properties deserve emphasis. First, latent-space models avoid the trap of judging a plan by how realistic its rendered future looks, because nothing is rendered. Second, planning cost is real: every candidate sequence requires a rollout, so these systems are usually slower per control step than a direct policy and their reported success rates should be read with that budget in mind. Neither of the two papers reports a controlled comparison against a well-tuned behaviour-cloning baseline trained on the same data, which is exactly the comparison a newcomer would want.

7. Family D: world models as data factories and as training environments

The offline uses are where world models have most clearly changed practice, because they attack the cost of robot data rather than the cost of inference.

Data generation (D1). DreamGen trains an action-conditioned video model on robot and general video, then labels the synthetic futures with an inverse-dynamics model to recover actions, producing demonstration data for new behaviours. The authors report scaling a single-task policy to 22 behaviours on a humanoid, and introduce a benchmark for evaluating generated robot video [NVIDIA, 2025b]. Cosmos supplies the general-purpose video world-model platform with open weights that this style of work builds on, though its published evidence is about generation quality and physical plausibility rather than downstream control [NVIDIA, 2025a]. PlayWorld, in the 2026 slice, argues from unsupervised play data that diverse contact-rich experience can be collected without task

labels; this report used only its abstract-level claims, so it is not counted as control evidence here [PlayWorld, 2026]. A substantial full-text HTML acquisition of that paper is retained in our evidence bundle, but it was acquired rather than read for the body of this report, so no claims here rest on its full text.

Training inside the model (D2). Dreamer 4 trains a policy entirely inside a learned world model, using imagined rollouts for reinforcement learning, and reports solving Minecraft from offline data without environment interaction [Hafner et al., 2025]. World-Gymnast applies the recipe to a real arm: a world model pretrained on large-scale robot video, then policy improvement by reinforcement learning inside the model, evaluated on a physical arm on 22 tasks. The reported comparison in that paper is 34% against 58% for the same policy trained with direct real-world reinforcement learning, and a 28.5% figure in unseen rooms; the authors present these as evidence that the model-based route is cheaper per unit of real interaction rather than uniformly better [World-Gymnast, 2026].

The central methodological hazard of this family is model exploitation: an optimiser that is allowed to search inside a learned model will find the model's errors, and an imagined success says nothing about the physical system. Papers in this family are aware of it and usually report real-robot confirmation, but the confirmation is typically small-sample and usually single-site, with the two-laboratory, two-arm planner demonstration of V-JEPA 2 the main exception in this corpus [Meta FAIR, 2025]. The second hazard is evaluation leakage: if the world model was pretrained on video that includes the test environment, an apparent generalisation result may be a retrieval result. Only the papers that explicitly hold out environments, as World-Gymnast does for its unseen-room number, give a reader any grip on this.

8. Data, embodiments and compute: the requirements are not symmetric

Table 2 summarises what each system needs. The asymmetry is stark. A competitive direct policy needs action-labelled robot data, which is expensive but well understood, and a fine-tuning recipe on a single GPU is now realistic at the small end: SmolVLA is explicitly designed for that budget [Shukor et al., 2025]. A world-model planner needs action-conditioned video of the same embodiment, which is harder to obtain than action labels in the open-source ecosystem, or a sufficiently good latent space that action-free video suffices [Zhou et al., 2024, Meta FAIR, 2025]. A world-model trainer needs all of that plus a reward or value signal, plus enough real-robot budget to validate that imagined improvement transfers [Hafner et al., 2025, World-Gymnast, 2026].

System	Prediction target	Data reported by the authors	Embodiment	Code / artefacts
OpenVLA [Kim et al., 2024]	discretised actions	~970k episodes, Open X-Embodiment	single Franka arm	open weights and fine-tuning code
pi-0 [Black et al., 2024]	action chunk (flow)	~10k hours, 7 configurations, 68 tasks	several mobile and static manipulators	weights released; training recipe partial
SmolVLA [Shukor et al., 2025]	action chunk	481 datasets, ~22.9k episodes, 10.6M frames	6-DoF arms, community hardware	code and weights, single-GPU training
GR00T N1 [NVIDIA, 2025c]	action chunk (diffusion)	data pyramid: web video, synthetic, real	humanoid	open weights and evaluation harness
Gemini Robotics 1.5 [Gemini Robotics Team, Google DeepMind, 2025]	action plus embodied reasoning	not disclosed in the paper	several (motion transfer)	not released
WorldVLA [Cen et al., 2025]	actions and next frame	LIBERO demonstrations	simulated arm	research code announced
DINO-WM [Zhou et al., 2024]	latent future	offline play data per environment	simulated and real arms	code released

System	Prediction target	Data reported by the authors	Embodiment	Code / artefacts
V-JEPA 2-AC [Meta FAIR, 2025]	latent future	<62 h unlabelled DROID video for the action module	two Franka setups, two labs	weights and code released
DreamGen [NVIDIA, 2025b]	video futures plus IDM actions	robot video plus general video	humanoid	model and benchmark described
Cosmos [NVIDIA, 2025a]	video	large video corpora	none (platform)	open weights, permissive licence
Dreamer 4 [Hafner et al., 2025]	latent state, reward, value	offline Minecraft data	game agent	code released
World-Gymnast [World-Gymnast, 2026]	latent future plus value	Bridge-style video, then real RL	one real arm	not released at time of writing

Table 2. Requirements as reported by the authors. Entries are not commensurable: "episodes", "hours" and "frames" count different things, and a humanoid corpus cannot be compared with a table-top arm corpus.

Two practical notes for a group without a robot fleet. First, several of these systems can be studied end to end in simulation with public assets: LIBERO provides 130 language-conditioned tasks in four suites and is the de facto comparison bed for the action-policy papers here [Liu et al., 2023]. Second, the world-model papers that are cheapest to reproduce are those whose dynamics are learned from video already on disk, because that removes the data-collection bottleneck entirely; DINO-WM's dependence on offline play data is a good example of a design that a small group can match [Zhou et al., 2024].

9. What can and cannot be compared

Comparable, with care:

- Same model, one factor changed. WorldVLA against its own action-only ablation is a legitimate comparison, because the data, architecture and evaluation are held fixed [Cen et al., 2025]. Ablations of this kind are the only place where "prediction helps" is a controlled claim.
- Same benchmark, same protocol. LIBERO numbers from different papers are often quoted side by side. They are only comparable when the suite, the observation set, the action space and the number of demonstrations match; the benchmark's own paper documents how much those choices matter [Liu et al., 2023].
- Rankings, not scores. RoboArena's contribution is precisely that pairwise preferences aggregate more reliably than absolute success rates across heterogeneous policies [RoboArena, 2025].

Not comparable, although frequently compared:

- Absolute success rates across embodiments. A humanoid task set and a table-top arm task set do not share a difficulty scale, and the denominators differ.
- Simulation against real. A LIBERO score of one family and a real-arm score of another are two different measurements of two different systems; the gap between them is the sim-to-real problem, not a property of a method.
- Predicted-video quality against control success. FVD and similar metrics measure distributional similarity of generated frames, and no paper in this corpus establishes a monotone relation between them and task success. A convincing rollout video is therefore an illustration, never a control result.
- Data-efficiency claims expressed in "times fewer samples", because the baselines and the counting units differ between studies [LBM blind-trial study, 2025, Zhou et al., 2024].

Figure 2 makes the resulting shape visible. Only the direct-policy column contains multi-institution evaluation at the level of a whole study, where several groups run one shared protocol and the results can be pooled: RoboArena's distributed arena and the blind comparison of large behaviour models [RoboArena, 2025, LBM blind-trial study, 2025]. The planner column is not uniformly single-site either: V-JEPA 2's action-conditioned planner was demonstrated on two robot arms in two laboratories [Meta FAIR, 2025]. That is a two-laboratory

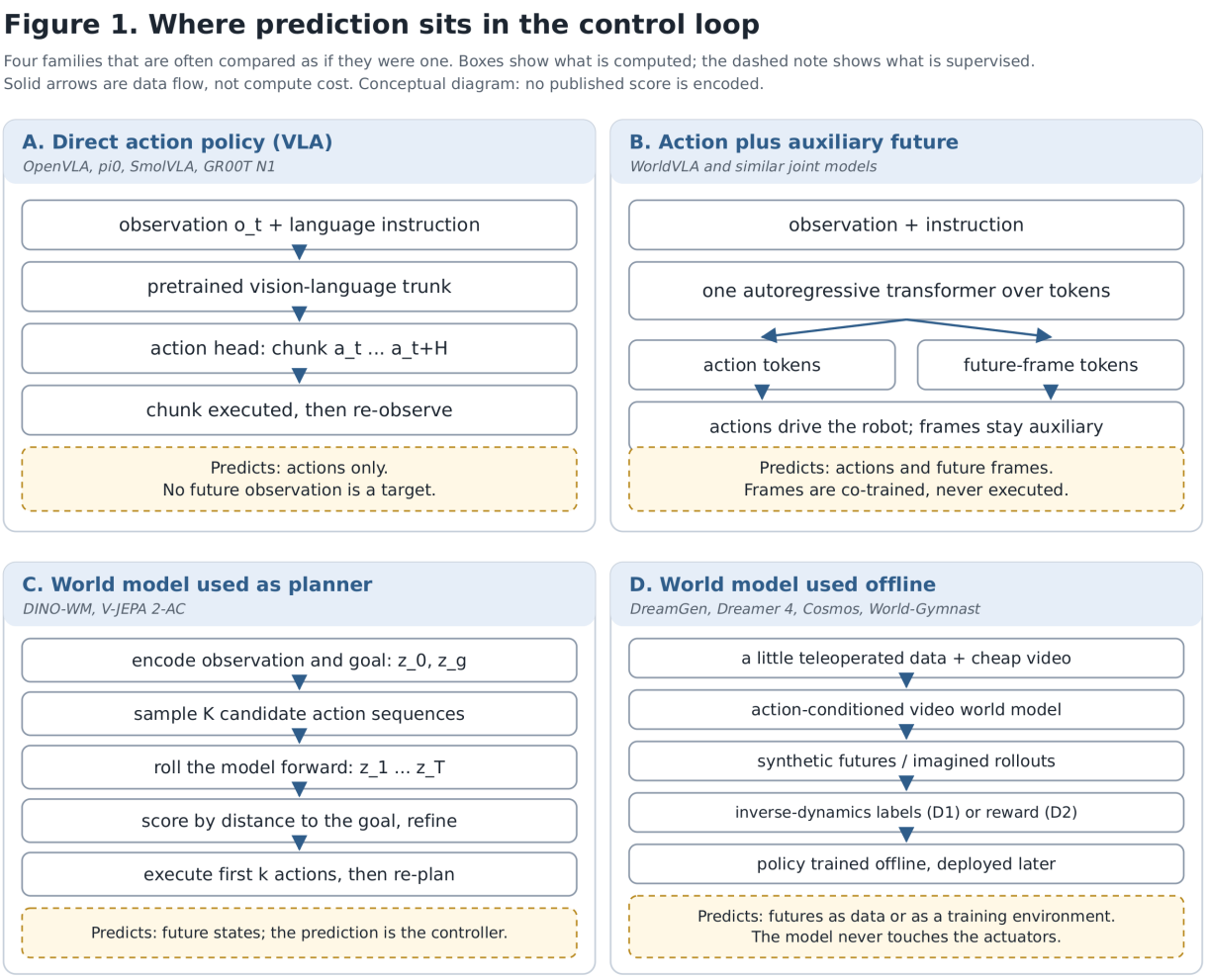
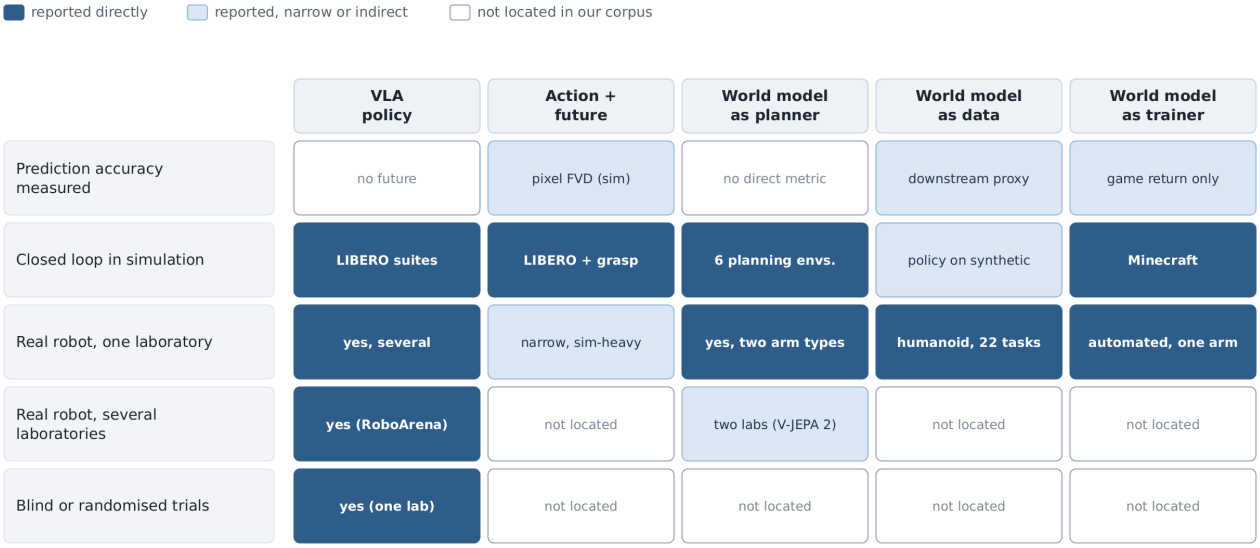


Figure 1: Where prediction sits in the control loop. Four families are drawn as data flow; the dashed note in each panel states what is supervised. The diagram is conceptual and encodes no published score.

Figure 2. What has been measured, by family

A reading of the sixteen papers cited in this report, not a leaderboard. Cells are qualitative; they carry no score and no cross-row comparison.



An empty cell means that we did not locate such a measurement in the sixteen papers we read; it is not evidence that no such measurement exists.
"Simulation" and "real robot" cells describe protocol breadth, not success rates: no success number is comparable across rows or columns.

Prepared for this survey from the cited papers. Conceptual reading aid.

Figure 2: A coarse evidence map over the sixteen papers. Cells record whether a measurement type was reported and how broad the protocol was; they carry no score. “Not located” means absent from this corpus, not absent from the literature.

demonstration of one system, which is weaker than a shared protocol that many groups can use to rank arbitrary policies, but it is more than a single-site result and it is recorded as such. The trainer column's closed-loop evidence is single-site, and in both the planner and trainer columns much of the supporting evidence is in simulation or in a game. This is the single most important asymmetry for a reader deciding where to spend effort, and it is a statement about evaluation breadth, not about which model is smarter.

10. Code, simulation and data resources that a small group can actually use

Open weights and training code are available for OpenVLA, SmolVLA, GR00T N1, V-JEPA 2 and Cosmos; DINO-WM and Dreamer 4 publish research code; WorldVLA announces code alongside its paper [Kim et al., 2024, Shukor et al., 2025, NVIDIA, 2025c, Meta FAIR, 2025, NVIDIA, 2025a, Zhou et al., 2024, Hafner et al., 2025, Cen et al., 2025]. pi-0 releases weights but not a full replication recipe [Black et al., 2024]; Gemini Robotics 1.5 releases neither [Gemini Robotics Team, Google DeepMind, 2025]; World-Gymnast had no public artefacts at the time of writing [World-Gymnast, 2026]. Licences differ, and Cosmos in particular carries a use policy rather than an ordinary open-source licence, which matters if the output is to be redistributed [NVIDIA, 2025a].

For simulation, LIBERO is the workhorse for language-conditioned manipulation and is used by most of the policy papers discussed here [Liu et al., 2023]. It is not a physics-fidelity benchmark: it is a task-diversity benchmark, and results in it saturate, which is why the more recent papers add real-robot confirmation rather than higher simulated scores.

11. Feasible next investigations without a fleet

The following are ordered by cost, and each is designed so that a negative result is still publishable inside a group.

P1. Reproduce one action-only baseline and one auxiliary-future variant on a single GPU. Take an open VLA checkpoint, fine-tune it on a small LIBERO suite, then add a next-frame prediction loss and re-run. The question is whether the reported auxiliary-prediction gain survives a different base model [Cen et al., 2025, Liu et al., 2023]. Cost: GPU hours, no hardware.

P2. Measure whether world-model fidelity predicts planning success. Train a latent dynamics model on an offline dataset, then plot planning success against held-out latent-prediction error as data scale increases. This tests the assumption that better models make better planners, which the literature asserts more often than it measures [Zhou et al., 2024, Meta FAIR, 2025]. Cost: single GPU, public video.

P3. Audit environment leakage in a published world-model result. For a paper that reports generalisation, take its pretraining corpus and test whether the held-out scenes are visually retrievable from it. This is a cheap, high-value check that no paper in this corpus performs systematically [World-Gymnast, 2026, NVIDIA, 2025b].

P4. Compare planning against behaviour cloning under an equal data budget. Same offline data, same embodiment, one planner and one direct policy, both evaluated on the same held-out tasks with a blind scoring script. This is the comparison the field lacks and the one that decides the practical question in Section 1.

P5. Replicate a multi-site protocol at small scale. RoboArena's lesson is that pairwise preferences aggregate better than absolute scores; a two-lab or two-operator version with a shared task list and a fixed comparison budget is within reach of a small consortium and directly addresses the weakest part of Figure 2 [RoboArena, 2025, LBM blind-trial study, 2025].

12. Limitations of this report

The corpus is sixteen papers, weighted towards 2024–2025, with a two-paper 2026 slice; it is not a current or comprehensive survey of the field, and it was assembled opportunistically rather than by a registered protocol. Of those two 2026 items, one (World-Gymnast) was read as a complete text, while the other (PlayWorld) is retained in the evidence bundle as a substantial full-text HTML acquisition from which this report used only abstract-level statements; what was acquired for that paper therefore exceeds what this report actually read and relies on. No number here was re-measured, no implementation was run, and no benchmark was re-scored, so every quantitative statement is an attributed claim under the original authors' conditions. Figures 1 and 2 are qualitative reading aids constructed for this report: they encode no score, and their "not located" cells record the limits of this corpus. The accompanying PDF was produced mechanically from this Markdown. Figure layout is

enforced geometrically in the figure generator, which aborts on any text overflow. The rendered figures and two PDF pages were inspected once with a vision model, whose only finding was duplicated figure-caption prefixes, since corrected; a later re-inspection attempt failed with a provider error and was not retried, so there has been no automated page-by-page visual sign-off of every page of the final PDF, and any such inspection would be a layout check rather than a scientific review of the content.

13. Recommendation

Build on a direct action policy first. It has the shortest path from public weights to a working system, it is where the evaluation methodology is most mature, and its failure modes are the ones a new group needs to learn to measure. Add prediction where it is cheap and testable: an auxiliary future-prediction loss costs one extra decoder and one ablation, and that is the experiment that tells you whether prediction helps *your* data. Treat world models as planners or trainers only once an offline data pipeline and a scoring harness exist, because both uses multiply the cost of every mistake in the data pipeline. And when reading the next wave of results, ask the question that separates the families: not how good the generated future looks, but what was measured, on how many sites, with what protocol.

References

Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Allen Z. Ren, Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and Ury Zhilinsky. pi-0: A vision-language-action flow model for general robot control. arXiv:2410.24164, 2024.

Jun Cen et al. Worldvla: Towards autoregressive action world model. arXiv:2506.21539, 2025.

Gemini Robotics Team, Google DeepMind. Gemini robotics 1.5: Pushing the frontier of generalist robots with advanced embodied reasoning, thinking, and motion transfer. arXiv:2510.03342, 2025.

Danijar Hafner et al. Training agents inside of scalable world models. arXiv:2509.24527, 2025.

Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. Openvla: An open-source vision-language-action model. arXiv:2406.09246, 2024. Conference on Robot Learning (CoRL).

LBM blind-trial study. A careful examination of large behavior models for multitask dexterous manipulation. arXiv:2507.05331, 2025. Preprint reporting a large-scale blind comparison of robot policies; cited by study name because no verified author list was retained in this task.

Bo Liu et al. Libero: Benchmarking knowledge transfer for lifelong robot learning. arXiv:2306.03310, 2023. NeurIPS Datasets and Benchmarks.

Meta FAIR. V-jepa 2: Self-supervised video models enable understanding, prediction and planning. arXiv:2506.09985, 2025.

NVIDIA. Cosmos world foundation model platform for physical ai. arXiv:2501.03575, 2025a.

NVIDIA. Dreamgen: Unlocking generalization in robot learning through video world models, 2025b. Preprint.

NVIDIA. Gr00t n1: An open foundation model for generalist humanoid robots. arXiv:2503.14734, 2025c.

PlayWorld. Playworld: unsupervised play as a source of robot manipulation data, 2026. Preprint; a full-text HTML acquisition (arXiv:2603.09030) is retained in this task's evidence bundle, but the report used only abstract-level statements from it, and no author list was verified.

RoboArena. Roboarena: Distributed real-world evaluation of generalist robot policies. arXiv:2506.18123, 2025. Preprint; cited by project name. The acquired full text did not supply a verified author list in this task, so no personal author names are asserted.

Mustafa Shukor et al. Smolvla: A vision-language-action model for affordable and efficient robotics, 2025. Preprint.

World-Gymnast. World-gymnast: reinforcement learning for a real robot inside a learned world model, 2026. Preprint; cited by project name because the author list was not verified from the acquired record in this task.

Gaoyue Zhou et al. Dino-wm: World models on pre-trained visual features enable zero-shot planning. arXiv:2411.04983, 2024.